SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

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SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

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1. Team & Project Overview

Team Name	Falcons
Project / Solution Name	KiraOrder for Orbyte
Name (Leader)	ZHOU JINGXI
Email & Phone (Leader)	DSC2509065@xmu.edu.my | +60 174842902
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	Primary: https://hozier123.github.io/sdc-2026-team-project/ Legacy / old UI: https://kiraorder-web.replit.app API: https://kiraorder-api.replit.app/api/v1 Owner reports RunPod adapter with fallback; Local Demo uses synthetic data.
Code Repository URL (public / access granted to judges)	https://github.com/Hozier123/sdc-2026-team-project

2. Architecture & Technology Stack

2.1 System Architecture Overview

KiraOrder helps clerks turn pasted multilingual order messages into reviewable drafts, owner decisions and delivery orders. React/TypeScript calls FastAPI on Replit; Neon PostgreSQL stores workflow records. A configured model branch sends masked text to Qwen2.5-7B-Instruct-AWQ through vLLM on RunPod. Exact aliases can resolve without AI. Tenant rules validate catalogue matches, calculate prices and check stock; humans approve orders.
He Minghao reports AI_PROVIDER=runpod_adapter with RunPod Active=0 and public parse falling back; this is not an integrated live-AI demonstration. Local Demo is a separate browser-only synthetic-data mode. RunPod was exercised independently for extraction and latency evaluation. The architecture diagram in Appendix A distinguishes these paths.
This declaration covers runtime dependencies, development tools, synthetic datasets, cloud services and sponsor resources, based on repository records and member declarations reviewed on 25 October 2026. Unrecorded tool versions and unquantified usage are identified explicitly.

2.2 Languages, Frameworks & Libraries

Name & Version	Purpose in Project	License
Python 3.13; Node.js 22	Backend and frontend runtimes	PSF License; MIT
PostgreSQL 17.11	Relational database	PostgreSQL License
FastAPI 0.158.1; Pydantic 2.14.1; pydantic-settings 2.16.1	API, schema validation and settings	MIT
Uvicorn 0.54.0; Starlette 1.9.0	ASGI server and API infrastructure	BSD-3-Clause
HTTPX 0.28.1	HTTP client for model requests	BSD-3-Clause
psycopg 3.3.6	PostgreSQL driver	LGPL-3.0
pglast 7.21	Runtime dependency imported by app.migrations and app.seed for SQL parsing	GPL-3.0-or-later
RapidFuzz 3.14.6	Alias and catalogue matching	MIT
pytest 9.1.1; Ruff 0.17.1; jsonschema 4.26.0; PyYAML 6.0.3; Hatchling 1.32.4	Testing, linting, validation and build tooling	MIT
@gsap/react 2.1.2	React animation integration	GSAP Standard License
@react-three/drei 10.7.9	Three.js scene helpers	MIT
@react-three/fiber 9.8.2	React renderer for Three.js	MIT
gsap 3.15.0	Interface animation	GSAP Standard License

Name & Version	Purpose in Project	License
lucide-react 0.468.0	Interface icons	ISC
react 19.1.0	User interface	MIT
react-dom 19.1.0	Browser rendering	MIT
three 0.186.1	3D visual rendering	MIT
@tailwindcss/vite 4.3.3	Tailwind build integration	MIT
@types/react 19.1.0	React type declarations	MIT
@types/react-dom 19.1.0	DOM type declarations	MIT
@types/three 0.186.0	Three.js type declarations	MIT
@vitejs/plugin-react 4.5.0	React build integration	MIT
ajv 8.20.0	JSON schema validation	MIT
ajv-formats 3.0.1	Schema format validation	MIT
openapi-typescript 7.6.1	API type generation	MIT
tailwindcss 4.3.3	Interface styling	MIT
tsx 4.19.4	TypeScript tooling execution	MIT
typescript 5.8.3	Typed frontend code	Apache-2.0
vite 6.4.5	Frontend build and development	MIT
vLLM 0.32.0	RunPod model serving; version reported by operator	Apache-2.0
Marcellus; Cormorant Garamond; Cinzel	Self-hosted fonts; Cinzel retained in repository, not claimed as active UI font	SIL Open Font License 1.1
React Bits DepthText, TS-CSS; retrieved 23 October 2026	Opening wordmark depth effect; source hashes in frontend/src/components/reactbits/SOURCE.json	MIT + Commons Clause License Condition v1.0
Official SDC repository starter	Code scaffold	Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod OpenAI-compatible inference API https://lx22t9308hc10c.api.runpod.ai	Standalone evaluation and backend model path. Owner reports adapter configured; public parse currently falls back with Active=0.	SDC sponsor credit: US$25
Neon PostgreSQL connection	Backend database access through psycopg; a database protocol connection, not a GenAI API.	Free plan
External website assets and references	Frontend visual/font assets and design references are distinct from order-processing APIs. No WhatsApp, payment, mapping, GPT or Gemini API is claimed in the runtime order flow.	Asset-specific terms; no runtime GenAI subscription

3.2 Cloud Platforms & Deployment

Service & Provider	Purpose in Project	Plan / Region
Replit — backend	FastAPI deployment at kiraorder-api.replit.app; public deployment screenshot in Appendix B.	Core US$20/month, self-paid by He Minghao; Asia; Autoscale 2 vCPU / 4 GiB
Replit — frontend development and hosting	Legacy frontend / old UI at kiraorder-web.replit.app. Primary UI is GitHub Pages. Ye Lianghui used sponsored Agent assistance.	"Power" as reported by Ye; region not recorded; US$6.31 sponsored usage
Neon	PostgreSQL 17.11; project kiraorder-demo-2026-pg17.	Free; ap-southeast-1; organization Minghao
RunPod Serverless Load Balancer	vLLM inference; PRO 6000 MIG, 24 GB, one GPU. Container vllm/vllm-openai:latest; operator reports vLLM 0.32.0.	US-NE-1; US$0.00019/second; US$25 sponsored credit; screenshot shows zero workers
GitHub / GitHub Pages	Source control, review, evidence records and primary frontend at hozier123.github.io/sdc-2026-team-project/.	Plan not recorded; managed hosting region not specified

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Qwen2.5-7B-Instruct-AWQ 4-bit AWQ	Qwen / Hugging Face; served on RunPod	Pre-trained; no training or fine-tuning. Temperature 0; max model length 8192; max output tokens 2048. No FP16 comparison run is claimed.	Apache-2.0 https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-AWQ

4.2 Datasets

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	PII? Y/N
T1 seed	Team; Zhan's MB-T1-20261022-r1 and He's ID mapping	Private synthetic; 40 SKUs, 25 families, 8 customers	Team-owned synthetic data; no third-party license	N
T2 seed	Team-created tenant fixture	Private synthetic; 16 SKUs, 10 families, 5 customers	Team-owned synthetic data; no third-party license	N
DEV41	Team development fixtures	Synthetic development/evaluation set; not an independent frozen accuracy benchmark	Team-owned synthetic data; no third-party license	N
TEST32	Team-controlled inputs and answer key; He: structure/fields/tenant mapping review; Zhan: business review; Zhou: evaluation package and freeze/scoring records	Private synthetic evaluation; 32 items / 34 executions; 19 AI-only, 1 resolver-only, 12 workflow	Team-owned synthetic data; answers privately controlled	N
Frontend demo orders	Ye Lianghui / team	Synthetic UI demonstration data	Team-owned synthetic data; no third-party license	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	PII? Y/N
Canonical six-case samples	Team; docs/ai_contract/samples	Synthetic contract cases: normal, multi_line, multi_candidate, ambiguous, condition, empty	Team-owned synthetic data; no third-party license	N
Manual baseline package	Team baseline protocol and blank recording sheet	Prepared but not executed; all inspected raw rows are not_started. The 210-second / 30% figures remain estimates.	Team-owned; no third-party license	N

4.3 AI-Assisted Development Tools

Yes. Versions and usage levels below follow the members' latest declarations. Joint "major" ratings describe combined tool use where no per-tool rating was supplied. Previously disclosed tools remain listed unless withdrawn; unconfirmed levels are marked explicitly. Development assistants are not runtime application models.

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Codex Desktop 26.1013.40934 — He Minghao	Code implementation, review and documentation	Major
DeepSeek Harness 44.7.0 — He Minghao; desktop nightly channel	Coding and review assistance	Major
Qoder Desktop 0.4.11 — Xiong Peicheng	Read-only repository review, hash checks, evidence and script review	Moderate
Antigravity 2.4.3 with "Gemini 3.8 Flash High" — Xiong Peicheng	Execution agent for code, scripts, tests and evidence generation	Major
DeepCoder — Xiong Peicheng; version not recorded	Previously disclosed subagent support for batch code and evaluation; absent from the new usage-level statement	Pending member confirmation
Replit Agent / "Power" — Ye Lianghui	Homepage generation, layout, styles and animation; previously disclosed sponsored use	US$6.31 usage; qualitative level not supplied
Codex with GPT6.1 — Ye Lianghui	Minor assistance with actual development; project records, documents, website tutorials and GitHub repository comparisons	Major overall tool use, as jointly reported; coding use remains minor

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Gemini 4 — Ye Lianghui; latest reported label	Researching project-related information; replaces the earlier "Argon" label in the member update	Major combined use reported with Codex; separate level not specified
Codex — Zhan Zhengrong; version not recorded	Previously disclosed project assistance; absent from the new usage-level statement	Pending member confirmation
ChatGPT with GPT-6 — Zhan Zhengrong	Part of the toolset for demo-video planning, scripts, English narration, AI animation storyboards and project analysis	Major overall toolset use, as reported
WorkBuddy with DeepSeek v4.1 Flash — Zhan Zhengrong	Part of the same video and project-materials toolset; individual task allocation not specified	Major overall toolset use, as reported
Jimeng AI with Seedance 2.0 mini — Zhan Zhengrong	A small amount of AI-generated content is used in the demonstration video. Asset filenames were not supplied. None is embedded in these two DOCX files.	Minor video contribution, confirmed by Zhan; overall toolset reported as major
Codex — Zhou Jingxi; version not recorded	Previously disclosed file organization and upload assistance; absent from the new usage-level statement	Pending member confirmation
Claude Code 2.1.178 — Zhou Jingxi	T1/T2 synthetic seeds, data scripts, SHA-256 checks, DEV/TEST materials, answer-key drafts, verification records and submission materials	Major

Human review and responsibility

Team members review generated code, documents and synthetic fixtures against the repository and business rules. Validation includes schema checks, hash manifests, backend tests and separate tenant-aware answer-key review. The model's output is not used to rewrite business truth to match a failing prediction. Failures and known limitations remain explicit. TEST32's approved freeze manifest fixes scope; 12 workflow cases remain NOT_RUN. We retain responsibility for final content, deployment behavior, licenses and human approval decisions. Personal tool versions and usage levels not recorded by members are not guessed.

5. Sponsor-Provided Resources

Yes. RunPod credits and Ye Lianghui's Replit Agent usage were sponsor-provided.

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
RunPod / SDC	US$25 cloud credit	GPU inference hosting and evaluation. Operator reports about US$11.6 used; screenshot balance US$13.38. These are reporting snapshots, not exact reconciled billing.
Replit / SDC	Sponsored Agent usage	Ye Lianghui reports US$6.31 for frontend generation and styling; activity 21 October, reporting period 20 October–19 November UTC. The supplied review and member declaration record a Usage screenshot showing US$6.31 Agent usage and total spend. The image is not in this local package; Usage is not an invoice or proof of sponsor origin.

Replaceability of sponsor resources

RunPod supplied the GPU service used for model evaluation and the configured extraction path. An equivalent model host with a compatible API could replace it. Replit supported frontend creation and backend hosting; the application could move to another suitable web host. The sponsor-supported frontend development account is separate from He Minghao's self-funded Replit Core deployment. Neon is a Free-plan service; no sponsor funding is claimed for it. No sponsor-provided dataset or additional sponsor GenAI API is claimed.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

Evaluation and demonstration use synthetic orders and catalogues, not real customer records. In connected operation, pasted order text reaches the backend; identifying text is masked before model processing and stored workflow use. Structured orders, approvals, delivery orders and audit records reside in Neon. Local Demo stores browser data; access tokens remain in tab memory. Role restrictions, tenant isolation and restricted warehouse fields limit access. The AI module does not persist raw messages; diagnostic evidence includes masked spans and timing. No fixed retention period or automatic deletion job is implemented. He Minghao handles test-database cleanup; Zhou Jingxi controls private evaluation files; Xiong Peicheng manages RunPod logs. Provider log retention is unconfirmed. RunPod processing occurs in US-NE-1 and Neon in Singapore. We do not claim a 30-day deletion policy or certified privacy compliance.

6.2 Known Limitations & Risks

He reports RunPod adapter configured, Active=0 and public parse falling back; this is not an integrated live-AI demonstration. The 23 October handoff records 198.5 seconds from scale-from-zero to first /health; the benchmark's 1.87 seconds was a hot-state first call. Model extraction can misread units: development evidence reports UOM 24/31 and six-case testing reports 5 PASS / 1 KNOWN_LIMITATION. The 23 live LLM cases with zero fallback have accuracy NOT_COMPUTED and are not an accuracy benchmark. TEST32 freeze records were approved and merged, but 12 workflow cases remain NOT_RUN. Closed-set family IDs, verbatim spans, schema validation, UNKNOWN handling and human decisions limit unsupported outputs. Price computation and approval remain outside the model. Remaining work includes live-path validation, workflow execution, unit handling and independent SME validation.

7. Compliance & Readiness Checklist

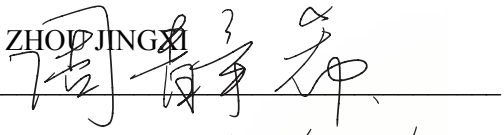
- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

Checklist items are marked on the basis of the team's confirmation. The leader's signature and signing date are intentionally left blank for separate signing.

8. Declaration

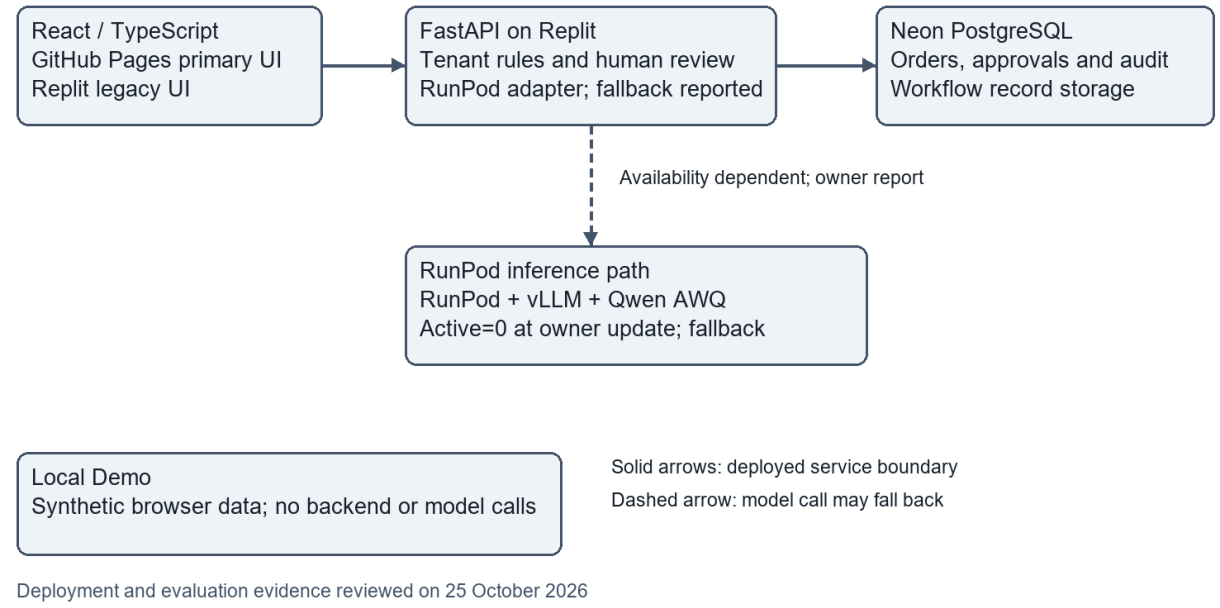
By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

Team leader: ZHOU JINGXI

Signature: 

Date signed: 2026/10/25

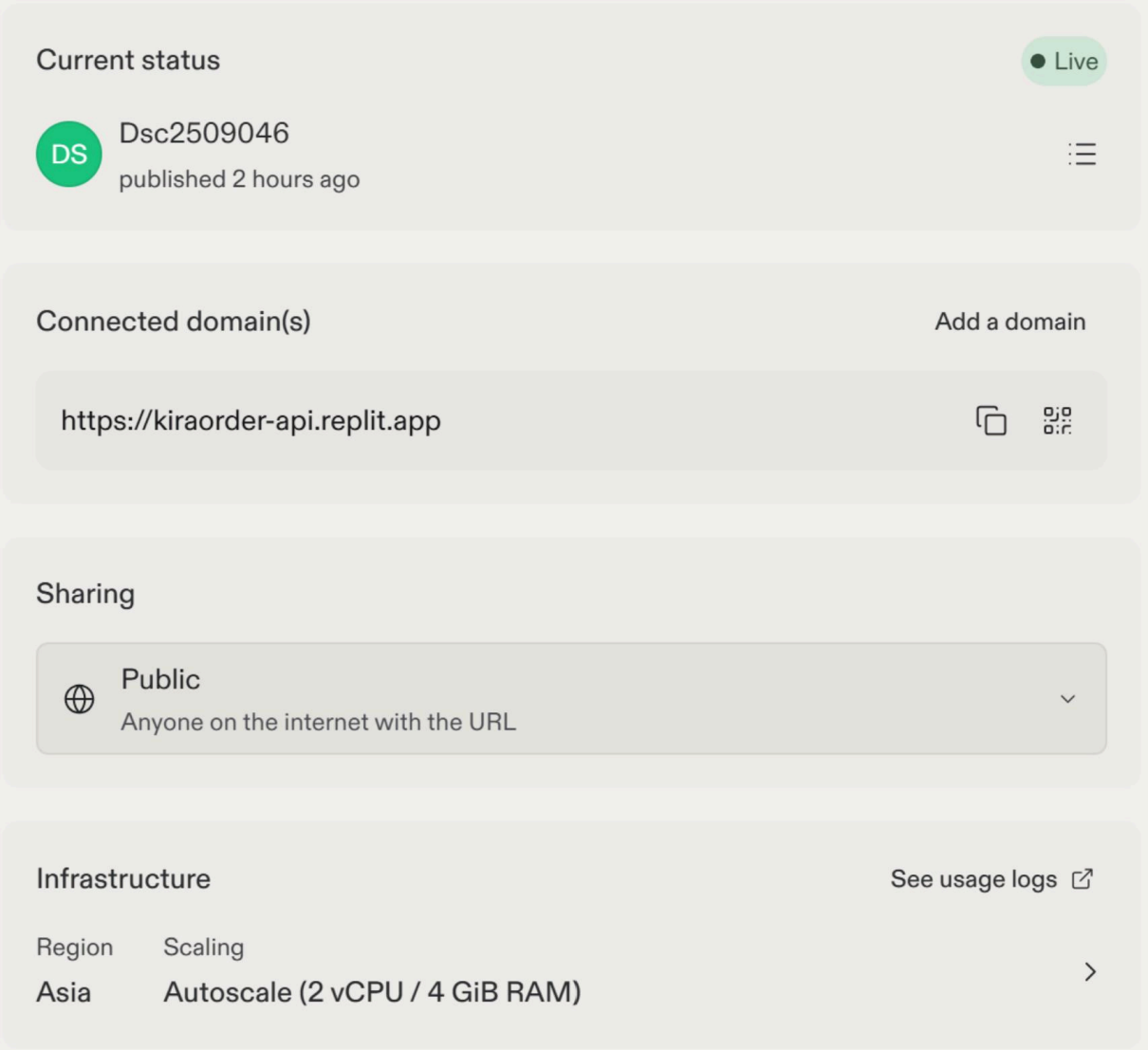
Appendix A Architecture and source records



https://github.com/Hozier123/sdc-2026-team-project/pull/65 — merged freeze manifest and approvals; 32 items / 34 executions, with workflow execution tracked separately.
https://github.com/Hozier123/sdc-2026-team-project/blob/main/docs/ops/stub-public-parse-gap.md — historical Stub configuration. He's latest supplied update reports RunPod adapter with fallback; no new live check is claimed.
https://github.com/Hozier123/sdc-2026-team-project/blob/main/evidence/EVIDENCE_INDEX.md — execution and limitation evidence.
https://github.com/Hozier123/sdc-2026-team-project/blob/main/frontend/package-lock.json and backend/requirements.lock — resolved dependency versions.
Model license: https://huggingface.co/Qwen/Qwen2.5-7B-Instruct-AWQ/blob/main/LICENSE
Other license references: https://github.com/encode/httpx/blob/master/LICENSE.md ; https://github.com/lelit/pglast ; https://gsap.com/standard-license
Cloud plans, sponsor amounts and personal development tools: member declarations provided by He Minghao, Xiong Peicheng, Ye Lianghui, Zhou Jingxi and Zhan Zhengrong. Unrecorded details remain explicitly marked. motionsite was a design/motion reference, not an evaluation or training dataset.

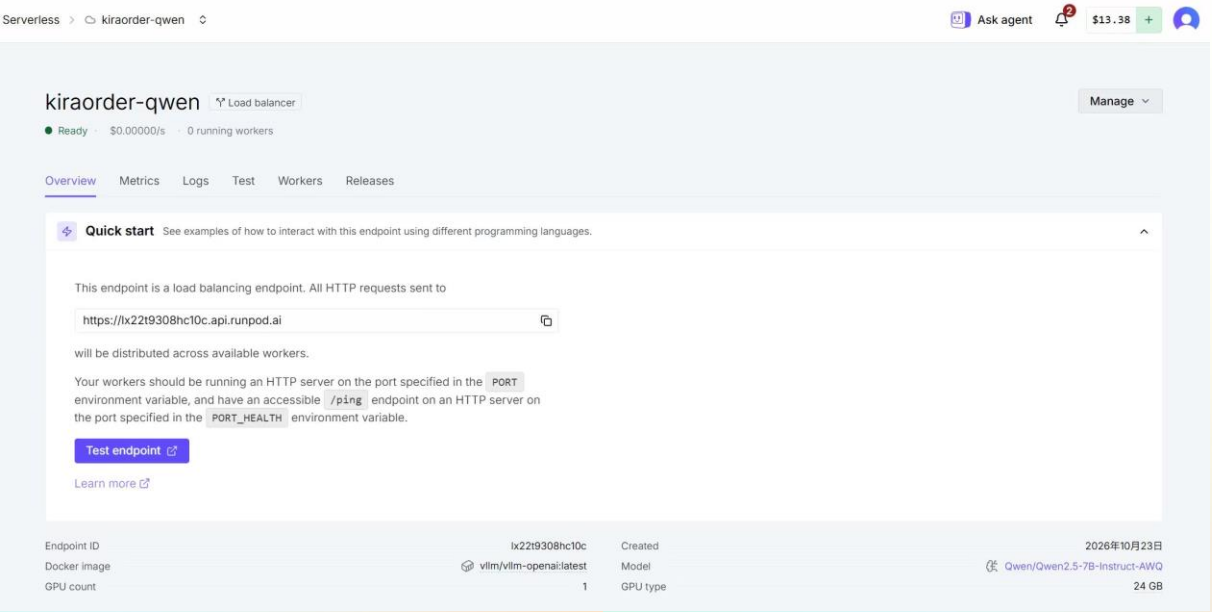
Appendix B Replit deployment evidence

Backend deployment screenshot supplied by He Minghao. It shows a live public deployment, the kiraorder-api.replit.app address, Asia region and Autoscale 2 vCPU / 4 GiB. This is hosting evidence, not proof of successful live model extraction; He's latest update reports the adapter configured but public parse falling back. The self-funded Core plan is declared separately from Ye Lianghui's sponsored Agent account.



Appendix C RunPod deployment evidence

Operator-supplied overview shows the serverless endpoint, GPU capacity, zero workers and a US$13.38 balance at capture. Zero workers is a snapshot after shutdown, not a claim of continuous availability. The endpoint was used for standalone evaluation. He now reports RunPod adapter configured, Active=0 and public parse falling back.

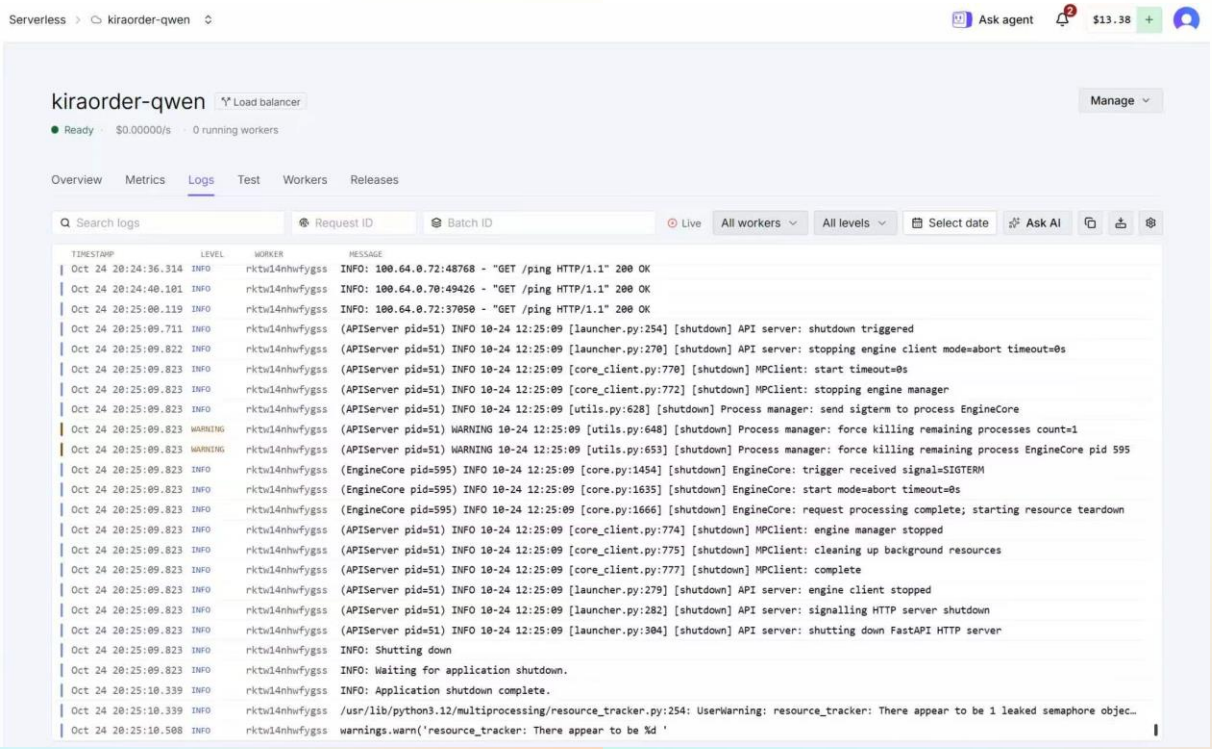


RunPod endpoint: https://lx22t9308hc10c.api.runpod.ai

Region: US-NE-1. Sponsored credit and usage are based on the operator’s declaration. No API key or bearer token is included.

Appendix D RunPod shutdown evidence

Operator-supplied logs record shutdown around 20:25:09–20:25:10 (+08:00) on 24 October 2026. The screenshot is retained as supplied, including the resource-tracker warning; shutdown logs do not establish inference accuracy or absence of all runtime warnings.



Appendix E Third party notices and source attribution

pglast 7.21

KiraOrder uses pglast under the GNU General Public License version 3 or later. It is a runtime dependency declared in backend/pyproject.toml and directly imported by app.migrations and app.seed; it is not merely a build tool. Upstream source: https://github.com/lelit/pglast (release v7.21). License text: https://www.gnu.org/licenses/gpl-3.0.html. This notice identifies the component and license; it does not assert that a public application repository alone fulfills every distribution obligation.

psycopg 3.3.6

KiraOrder uses the psycopg PostgreSQL driver under the GNU Lesser General Public License version 3. Upstream source: https://github.com/psycopg/psycopg (release 3.3.6). Upstream license: https://github.com/psycopg/psycopg/blob/master/LICENSE.txt. The LGPL incorporates GPL version 3 with additional permissions. Component notices and applicable source/redistribution requirements must be retained in distributed packages.

React Bits DepthText

Copyright (c) 2026 David Haz. The TS-CSS component was retrieved on 23 October 2026 and adapted for the opening wordmark. License: MIT + Commons Clause License Condition v1.0. The complete supplied notice is retained at frontend/src/components/reactbits/LICENSE-ReactBits.txt; source and file hashes are in SOURCE.json in the same directory. Registry: https://reactbits.dev/r/DepthText-TS-CSS.json. The component must not be described as plain MIT; the supplied license includes restrictions on distributing components themselves.

Fonts and scaffold

Marcellus, Cormorant Garamond and retained Cinzel are declared under SIL Open Font License 1.1. Cinzel is retained in the repository; current primary-interface use is not claimed. Cinzel attribution: Copyright 2020 The Cinzel Project Authors; notice at frontend/public/assets/fonts/Cinzel-OFL.txt. The official SDC repository starter is declared under Apache-2.0. Full dependency versions appear in Section 2.2.

Appendix F Visual assets and evidence boundaries

This inventory distinguishes verified code references from unresolved creator and permission records. A hosting URL does not establish ownership or authorization. motionsite is a design and motion reference, not training or evaluation data.

Asset and use	Recorded source	Creator or authorization status
Desktop background video and poster	PlanetGreeting.tsx: VIDEO_URL and POSTER_URL on d8j0ntlcm91z4.cloudfront.net; exact paths retained in source.	Creator / generating tool and permission evidence not supplied.
night-background-mobile.mp4 and night-background-mobile.jpg	frontend/public/assets; used as mobile background and poster.	Creator / generating tool and derivation rights not recorded in supplied evidence.
nebula.png	frontend/public/assets; frontend README describes extraction from the original intro HTML.	Original creator and permission evidence not supplied.
Marcellus and Cormorant Garamond	Self-hosted font assets and associated CSS.	SIL Open Font License 1.1; retain original font notices.
Cinzel	frontend/public/assets/cinzel.css and fonts/Cinzel-OFL.txt; retained resource.	Cinzel Project Authors; SIL Open Font License 1.1; active UI use not asserted.
Jimeng AI outputs	Zhan confirms a small amount of Jimeng AI / Seedance 2.0 mini content in the demonstration video; none is embedded in these two DOCX files.	Video use confirmed; individual asset filenames and a detailed asset list were not supplied.

Replit Usage evidence: Ye's declaration and the supplied comparison report record a screenshot showing US$6.31 Agent usage and total spend, with activity on 21 October in the 20 October–19 November 2026 UTC period. The actual Usage image is absent from the supplied local package and is not reproduced here. Sponsor origin is supported by Ye's explicit declaration, not by the Usage amount alone.

Deployment status: He's latest supplied update supersedes the earlier repository Stub note for configuration: AI_PROVIDER=runpod_adapter, RunPod Active=0, public parse falling back. Primary UI: GitHub Pages; Replit frontend is legacy. This revision did not perform a new authenticated parse test or certify current judge access.

Cold-start source: docs/handoff_2026-10-23.md, Section 1.2, reports 198.5 seconds after scale-from-zero to first /health. It explicitly identifies the benchmark's 1.87 seconds as a hot-worker first call. These are historical measurements, not current availability guarantees.